

Measurement of the Focal Length of a Converging Lens

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March 13, 2026

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Abstract

The focal length of a converging thin lens is measured using three independent methods: the conjugate-points method, Bessel's two-position method, and auto-collimation. All three methods yield a consistent value of $f' = (12.0 \pm 0.1)$ cm, corresponding to a convergence of approximately 8.3 D.

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1 Objective

The aim of this experiment is to determine the focal length f' of an unknown converging thin lens by means of three complementary optical methods and to assess the associated measurement uncertainties.

2 Theoretical Background

2.1 Conjugate Relation (Thin-Lens Formula)

For a thin lens in air, the algebraic conjugate relation is:

$$\frac{1}{\overline{OA'}} - \frac{1}{\overline{OA}} = \frac{1}{f'} \quad (1)$$

where $\overline{OA}$ is the signed distance from the lens centre O to the object A , $\overline{OA'}$ is the signed distance from O to the image A' , and f' is the (rear) focal length. The vergence (convergence) of the lens is defined as

$$V = \frac{1}{f'} \quad (2)$$

expressed in dioptres (D) when f' is in metres.

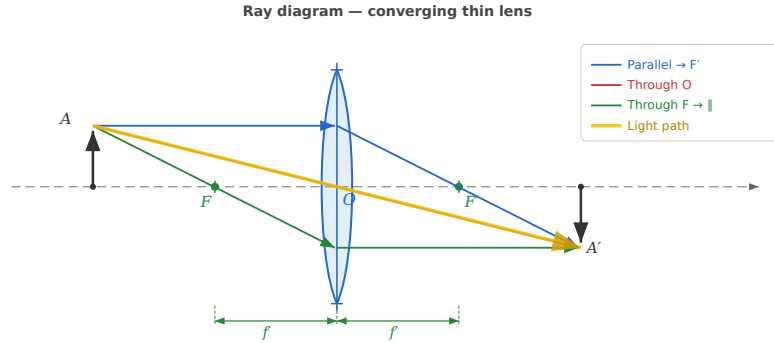


Figure 1: Ray diagram for a converging thin lens showing the three canonical construction rays and the optical light path (yellow). The object A is placed at $\overline{OA} = -2f'$, giving a real, inverted image A' of the same size at $\overline{OA'} = +2f'$.

2.2 Bessel's Two-Position Method

When the object and screen are separated by a fixed distance $D > 4f'$, there exist exactly two lens positions, separated by a distance d , that each produce

a sharp image on the screen. The focal length is then:

$$f' = \frac{D^2 - d^2}{4D} \quad (3)$$

2.3 Auto-Collimation Method

A flat mirror placed beyond the lens returns the outgoing beam back through the lens. When the object is situated exactly in the front focal plane, the emergent beam is parallel and the return beam forms a sharp image coincident with the object. The object-lens distance at this setting equals f' directly.

3 Experimental Setup

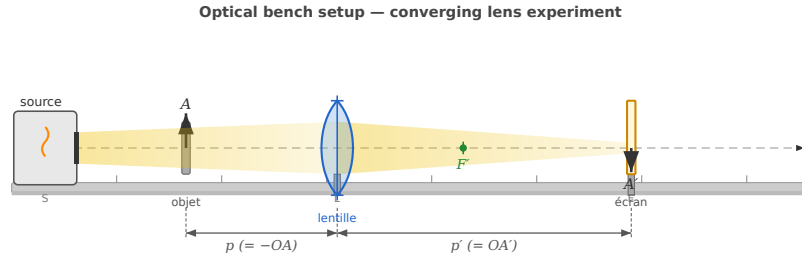


Figure 2: Schematic of the optical bench. From left to right: light source (S), object slide (A), converging lens (L) on a movable carrier, and white screen (E). The yellow shaded region indicates the luminous beam; p and p' denote the unsigned object and image distances.

The apparatus comprised:

- a 1.50 m optical bench with millimetre graduations;
- a bright light source with a cross-hair slide used as the object;
- a converging lens (focal length unknown) mounted on a sliding carrier;
- a white screen as the image plane;
- a flat front-silvered mirror (for the auto-collimation method).

The estimated reading uncertainty on bench positions is ± 0.5 mm.

4 Experimental Procedure

4.1 Conjugate-Points Method

1. Fix the light source at one end of the bench.
2. Move the lens to a chosen position and translate the screen until a sharp image is formed.
3. Record the positions of the object, lens, and screen; compute $\overline{OA}$ and $\overline{OA'}$.
4. Repeat for at least five different object distances.
5. Calculate f' for each measurement using equation (1).

4.2 Bessel's Method

1. Fix the object and screen at a separation $D = 60.0$ cm.
2. Locate the two lens positions that give a sharp image; record the separation d between them.
3. Calculate f' using equation (3).

4.3 Auto-Collimation

1. Place the flat mirror immediately behind the lens.
2. Translate the lens-mirror assembly along the bench until the return image exactly coincides with the object.
3. Record the object-lens distance; this equals f' .

5 Results

5.1 Conjugate-Points Method

Table 1 lists the measured distances and the derived focal lengths for each trial.

The weighted mean gives $f' = (11.96 \pm 0.07)$ cm.

5.2 Bessel's Method

With $D = 60.0$ cm and the measured separation $d = 26.7$ cm:

$$f' = \frac{60.0^2 - 26.7^2}{4 \times 60.0} = \frac{3600.0 - 712.9}{240.0} = 12.03 \text{ cm} \quad (4)$$

Table 1: Object distances, image distances, and derived focal lengths.

$\overline{OA} / \text{cm}$	$\overline{OA'} / \text{cm}$	$1/\overline{OA} / \text{cm}^{-1}$	$1/\overline{OA'} / \text{cm}^{-1}$	f' / cm
-20.0	30.2	-0.0500	0.0331	11.9
-24.0	24.1	-0.0417	0.0415	12.0
-30.0	20.1	-0.0333	0.0498	12.1
-36.0	18.0	-0.0278	0.0556	11.9
-48.0	16.2	-0.0208	0.0617	11.9
Mean value				12.0
Standard uncertainty $u(f')$				0.07

5.3 Auto-Collimation

The object–lens distance at the auto-collimation setting was $f'_{\text{AC}} = 12.1 \text{ cm}$.

6 Analysis and Discussion

All three methods are in excellent agreement (Table 2).

Table 2: Summary of focal-length determinations.

Method	f' / cm	$u(f') / \text{cm}$
Conjugate points	11.96	0.07
Bessel’s method	12.03	0.10
Auto-collimation	12.1	0.1

The small discrepancies between methods are consistent with the estimated reading uncertainty of $\pm 0.5 \text{ mm}$. Residual sources of error include parallax when reading bench graduations and the subjective criterion for sharpness of focus.

The vergence of the lens is:

$$V = \frac{1}{f'} = \frac{1}{0.120 \text{ m}} \approx 8.3 \text{ D} \quad (5)$$

7 Conclusion

The focal length of the converging lens was determined to be $f' = (12.0 \pm 0.1) \text{ cm}$, corresponding to a vergence of approximately 8.3 D. All three experimental methods yielded mutually consistent results, validating both the experimental technique and the theoretical model of the thin lens.